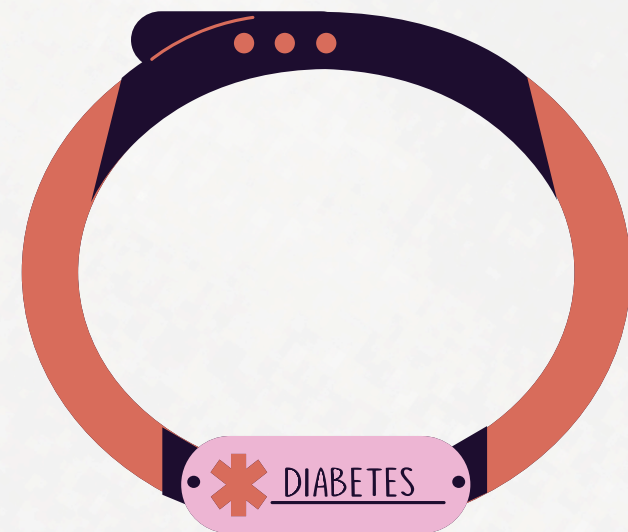


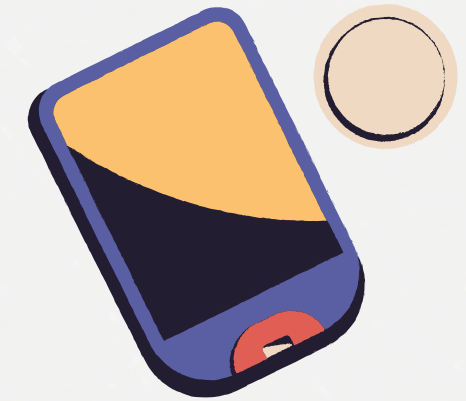
Symptom-based Diabetes Classification & Unsupervised Type Profiling



Dual-Stage Machine Learning Framework for
Early Clinical Triage

STUDENT
Filippo Velli





Problem Statement

Context

Diabetes → pervasive global challenge, causing severe long-term complications like neuropathy and cardiovascular disease

- **Type 1**, which requires acute intervention due to autoimmune beta-cell destruction
- **Type 2**, which is driven by progressive insulin resistance

Gap

Difficulty in **distinguishing** at the beginning which type a patient has

Risk of patients may not be funneled into the distinct therapeutic pathways required for their specific clinical presentation or prevention procedures

Objective

Correct identification of diabetic patients by using type specific signatures

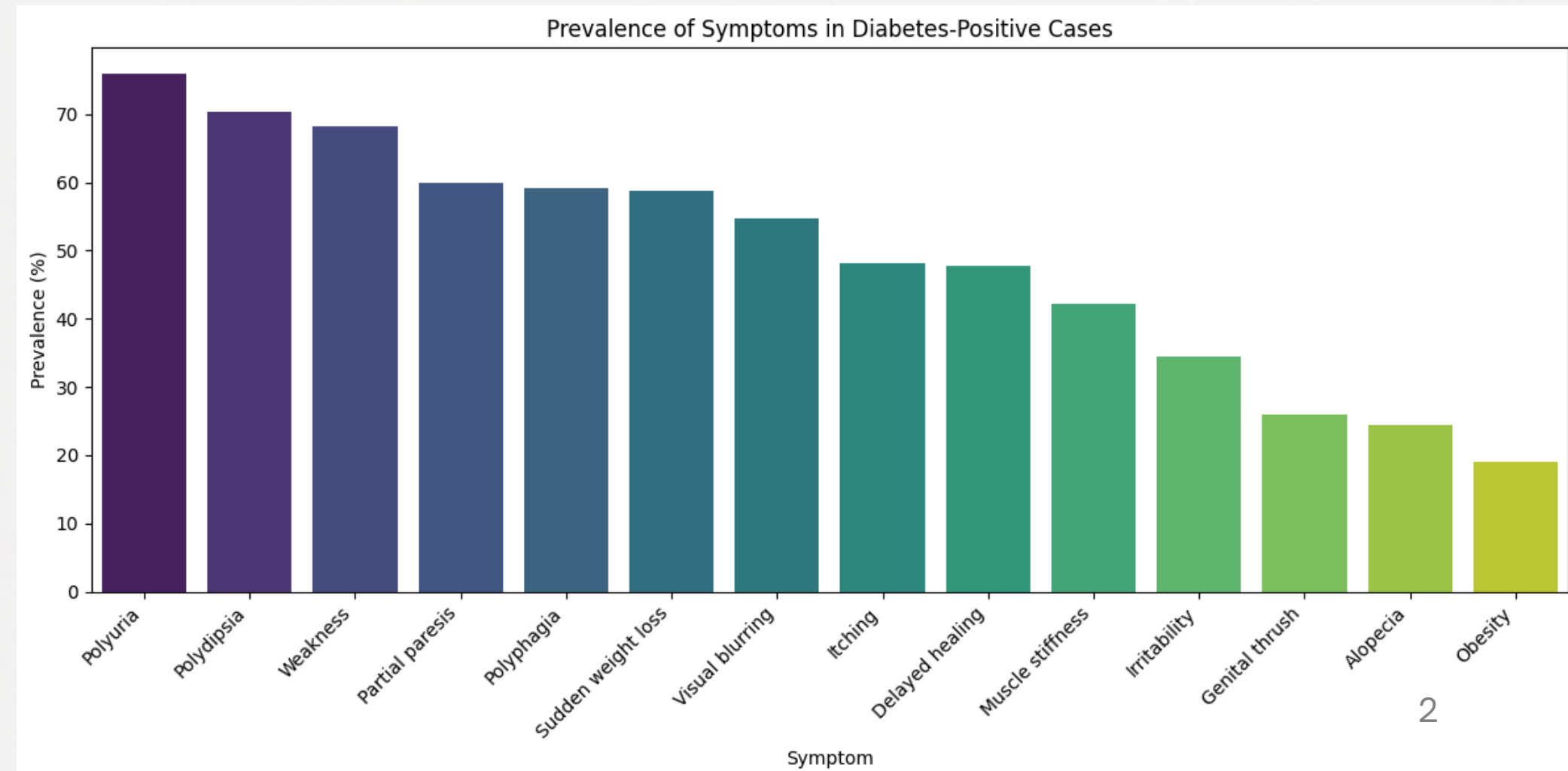
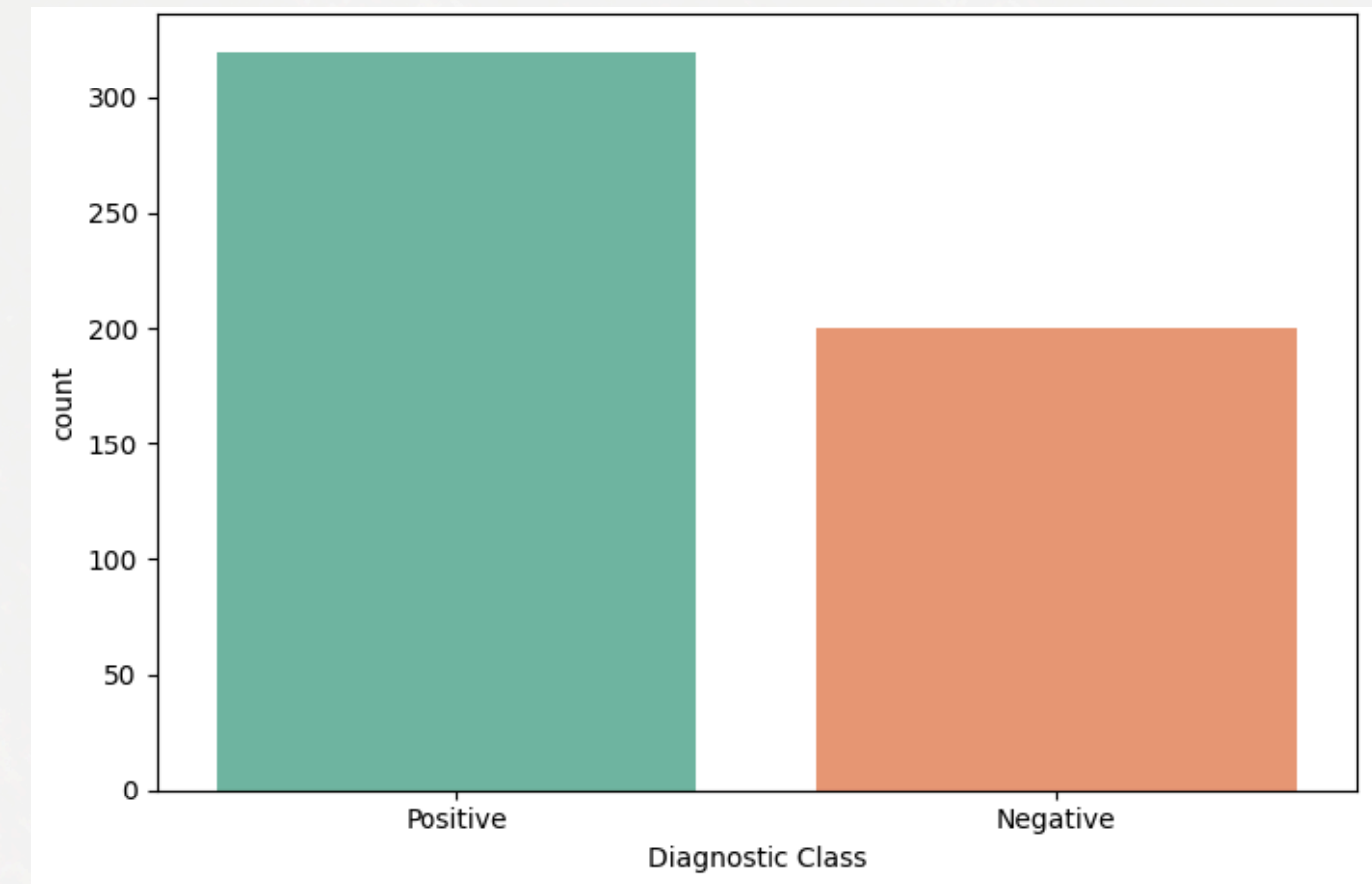
Ensuring more precise and timely clinical intervention and prevention

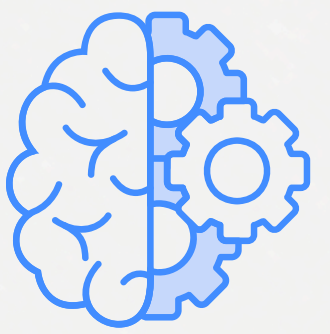
Dataset Overview

Note: Arsha Prasad made the dataset used in this work from data collected in the Sylhet Diabetes Hospital

Characteristics

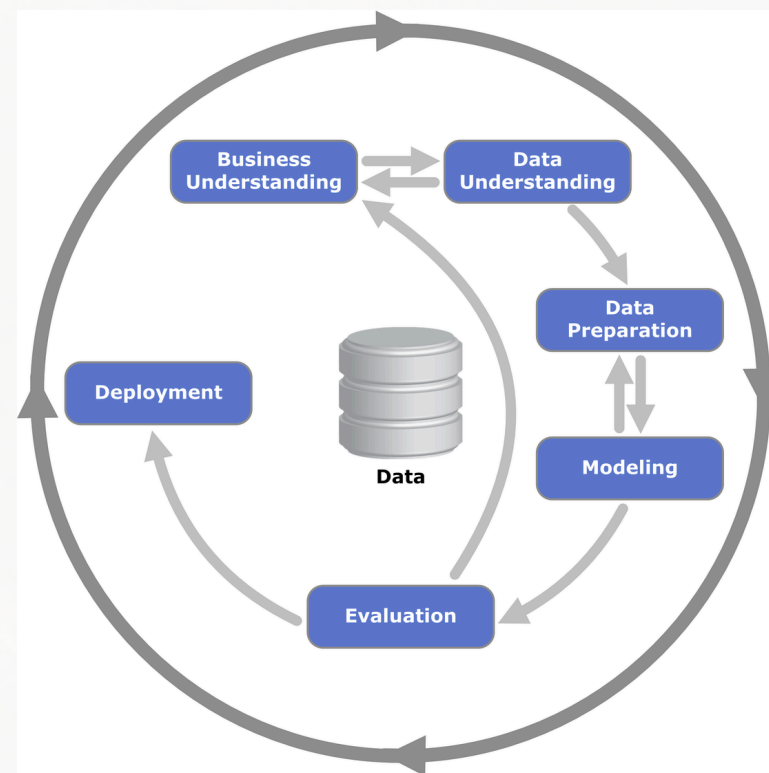
- 520 patients
- Binary target: Positive/Negative
- 62% positivity
- 16 features
- fully complete with no null values
- based on self-reports





Methodology: Dual-Stage Framework

CRISP-DM works as a starting point



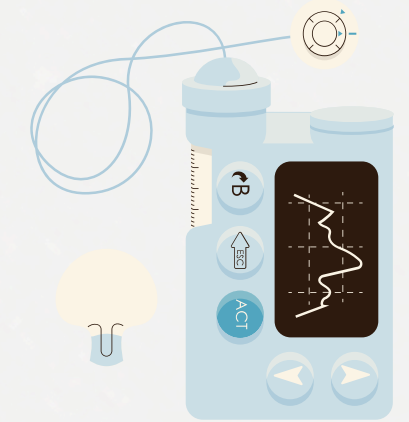
1st stage - Supervised classification

Establishment of an initial screening pipeline using ensemble methods to separate healthy subjects from the diabetic cohort accurately

2nd stage - Unsupervised clustering

Application of centroid-based clustering to the results of stage 1 to isolate patients affected by different types of diabetes

Data Understanding & Cleaning

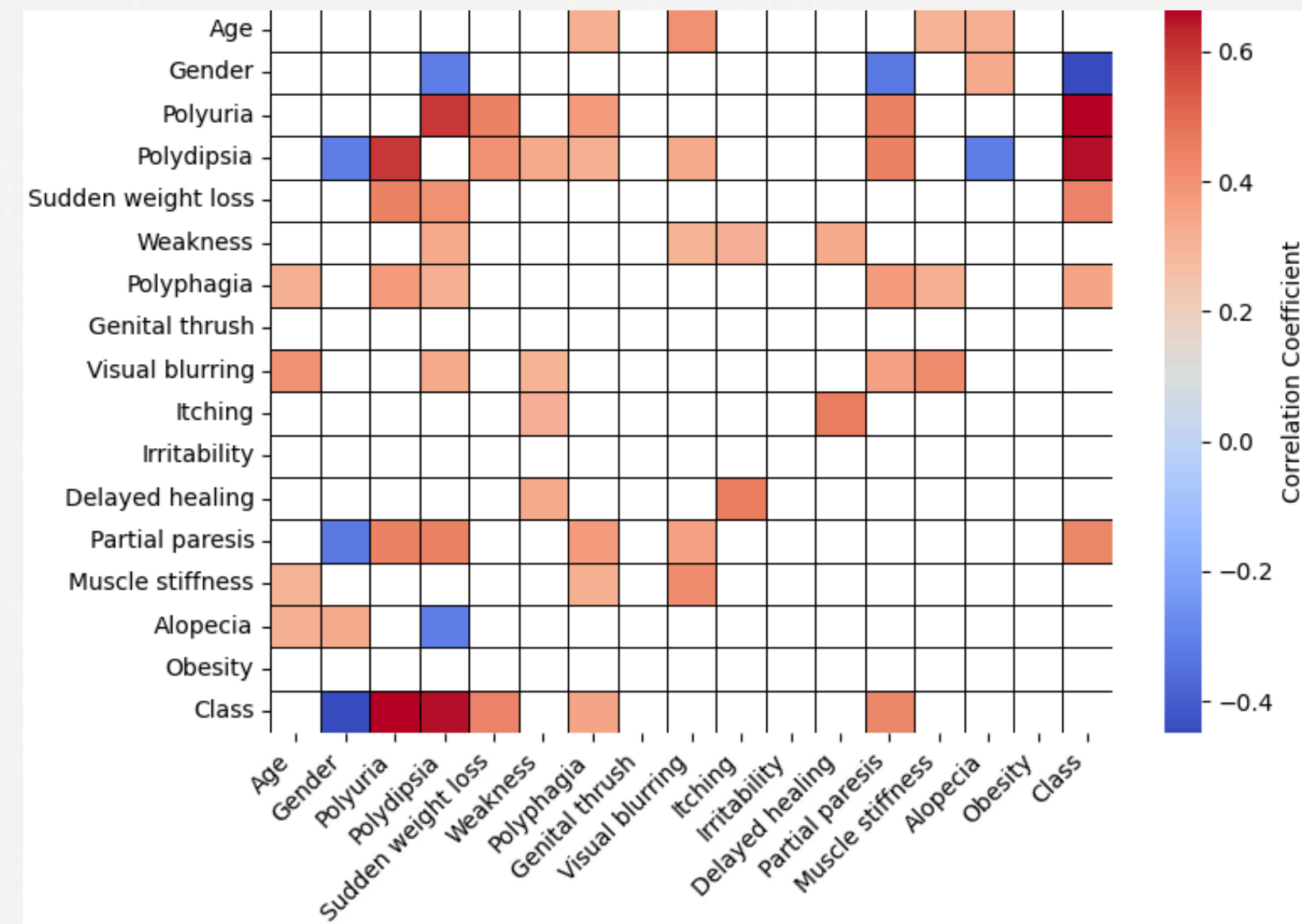


Preserving Clinical Diversity

- Retained **all outliers** for representativeness of atypical diagnostic cases
- Clustering phase stayed **sensitive** to unique patient signatures

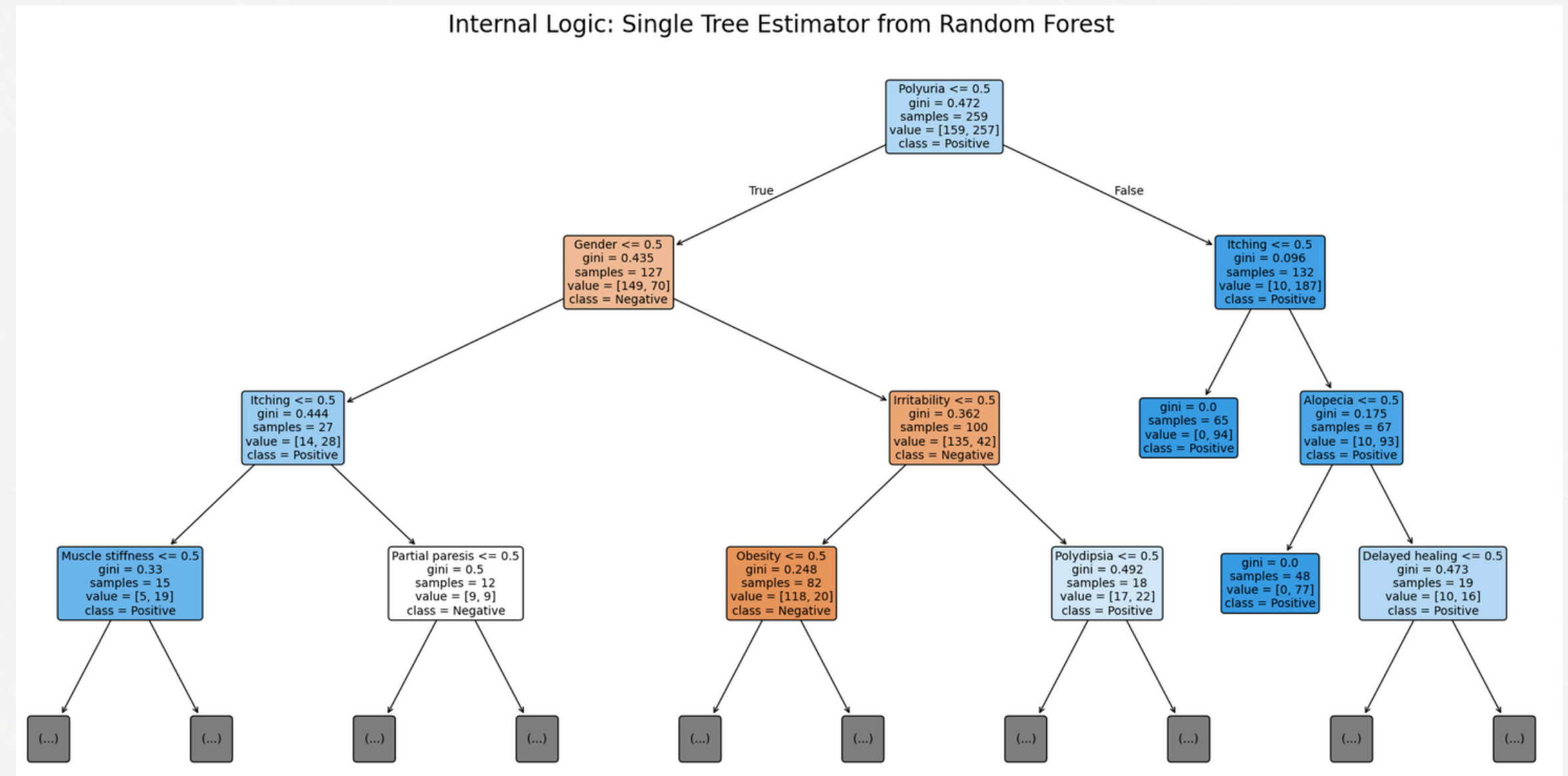
Statistical Integrity

- Implementation of a centralised **ColumnTransformer**: handle all categorical encoding and feature scaling
- **Leakage-free approach**: model's performance metrics reflect its true generalisation capabilities on unseen clinical data



Stage 1: Supervised Classification

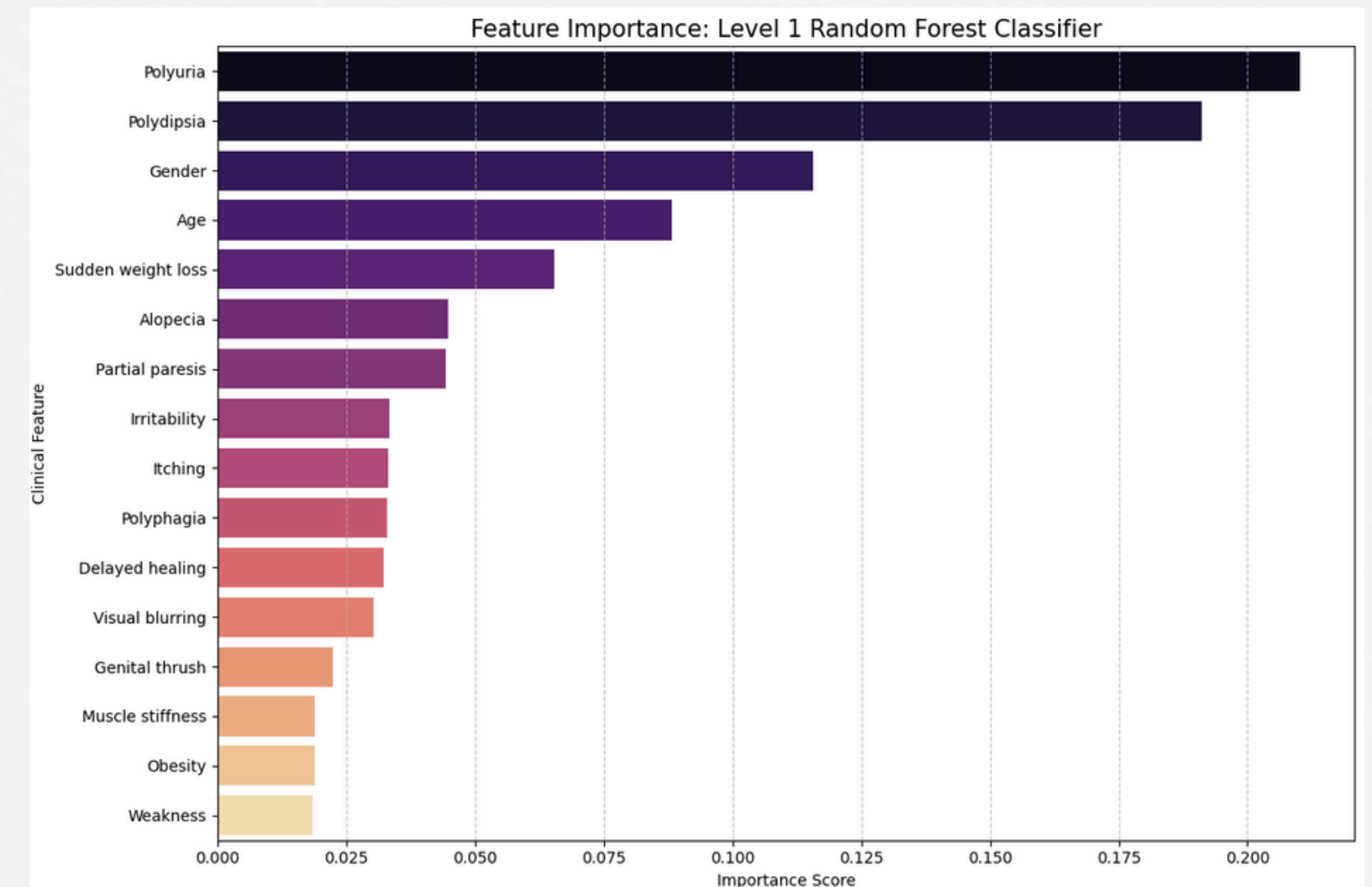
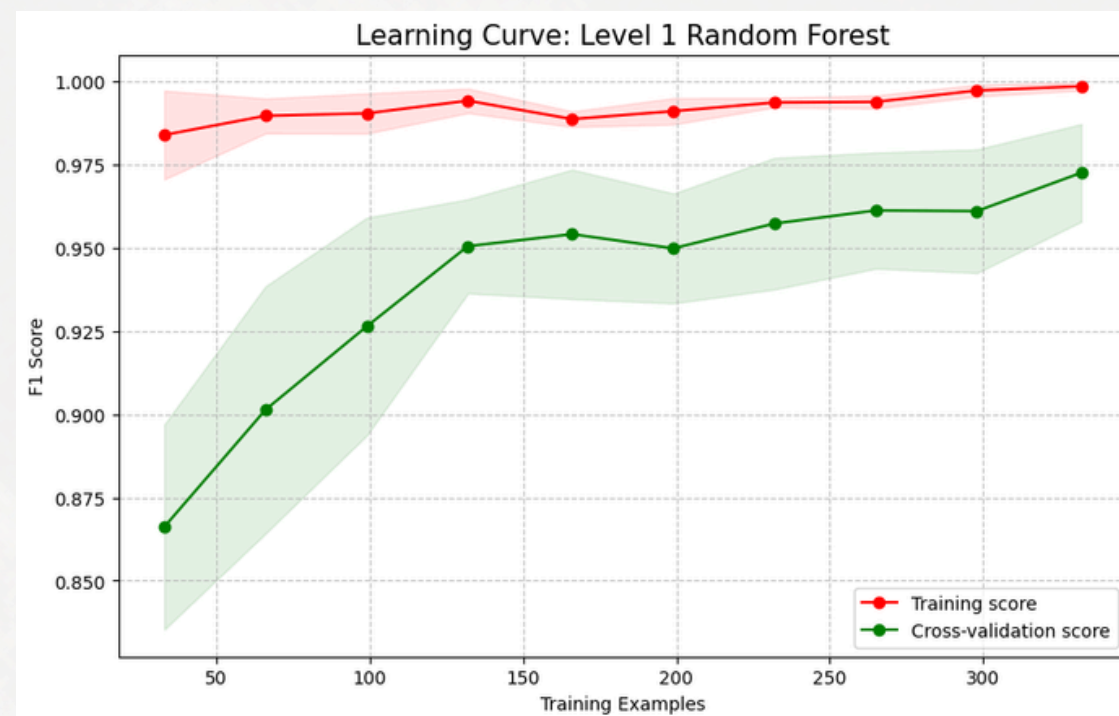
- **Random Forest** classifier was chosen as it can handle the complexity of the diagnostic screening phase
- Able to manage non-linear symptom interactions effectively, ensuring greater stability
- 5-fold cross-validation to ensure reliable model convergence across all subsets



Stage 1: Supervised Classification

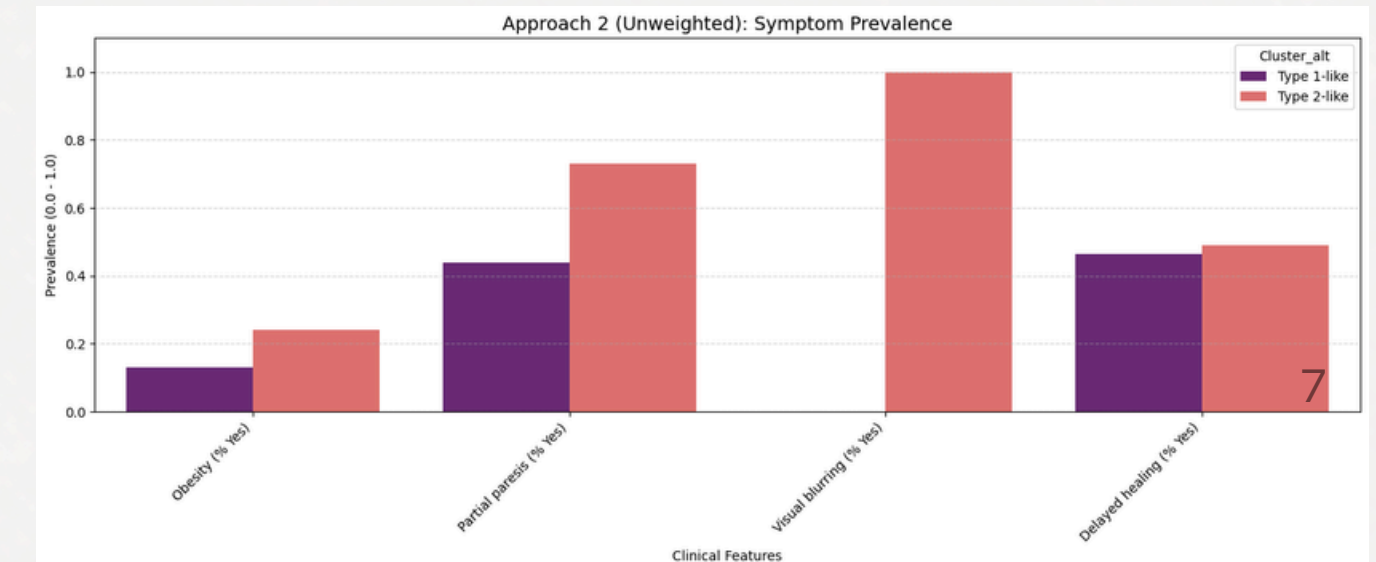
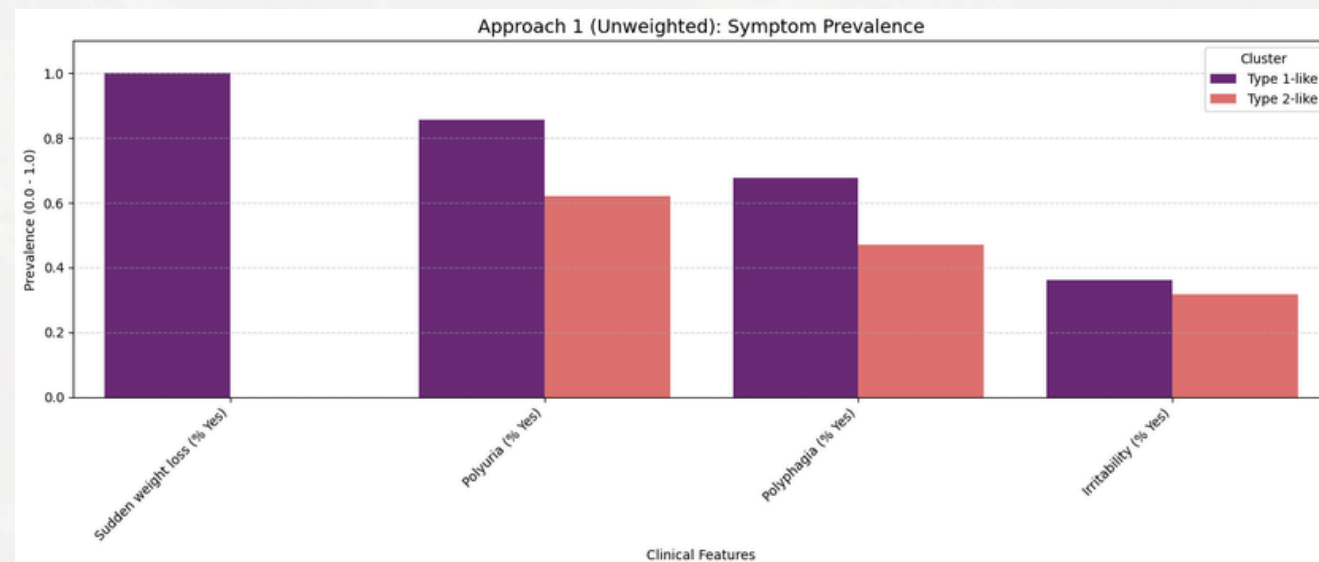
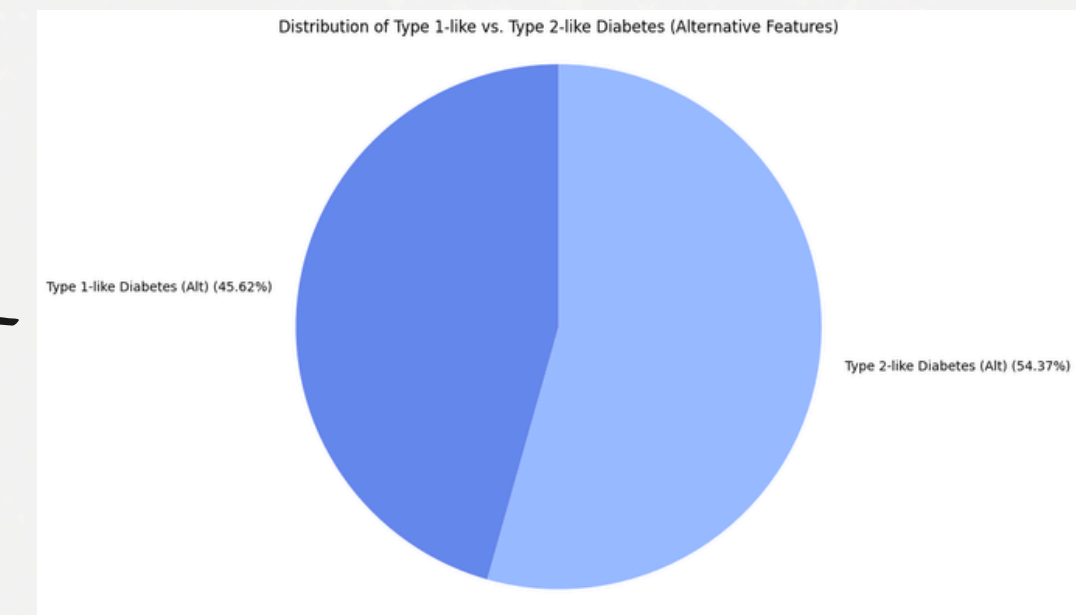
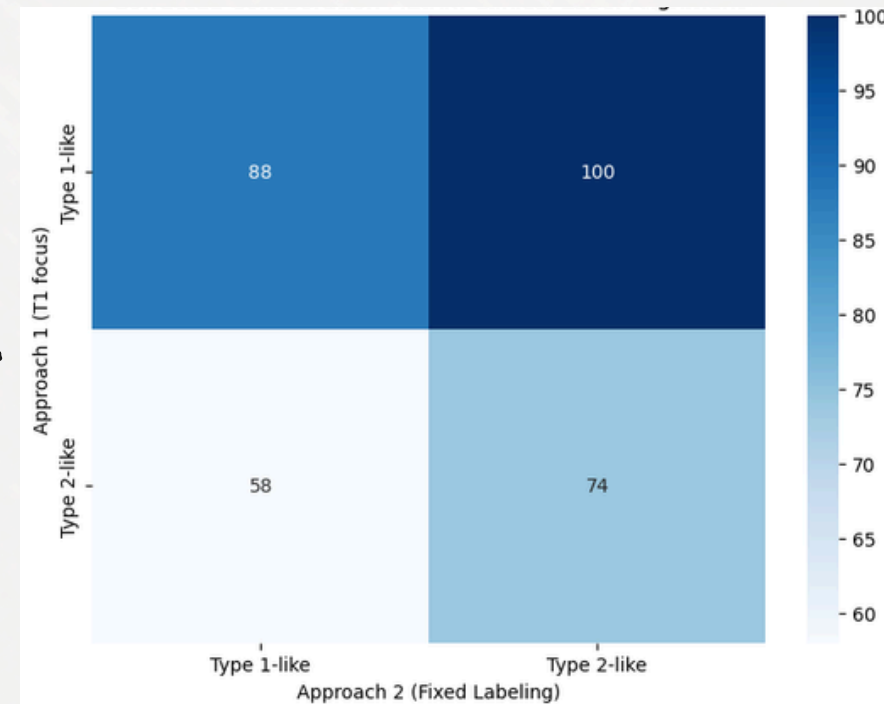
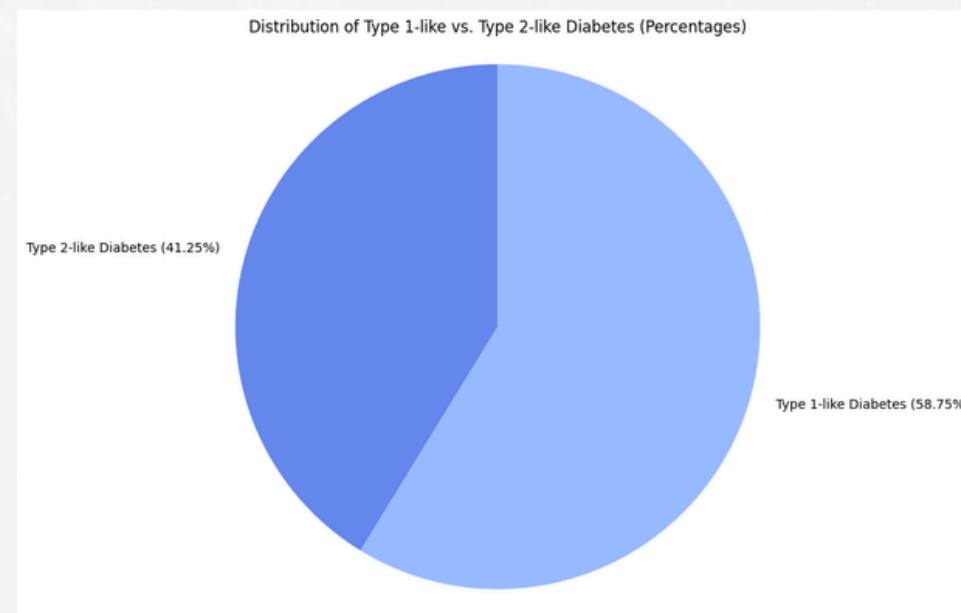
- Model achieved 97% overall accuracy, confirming the effectiveness of the screening pipeline
- The positive class reached an F1-score of 0.98, with minimal false negatives and false positives in the test partition

Model / Class	Precision	Recall	F_1 -Score
<i>Random Forest</i>			
Negative (Healthy)	0.97	0.97	0.96
Positive (Diabetic)	0.98	0.98	0.98
Accuracy			0.98



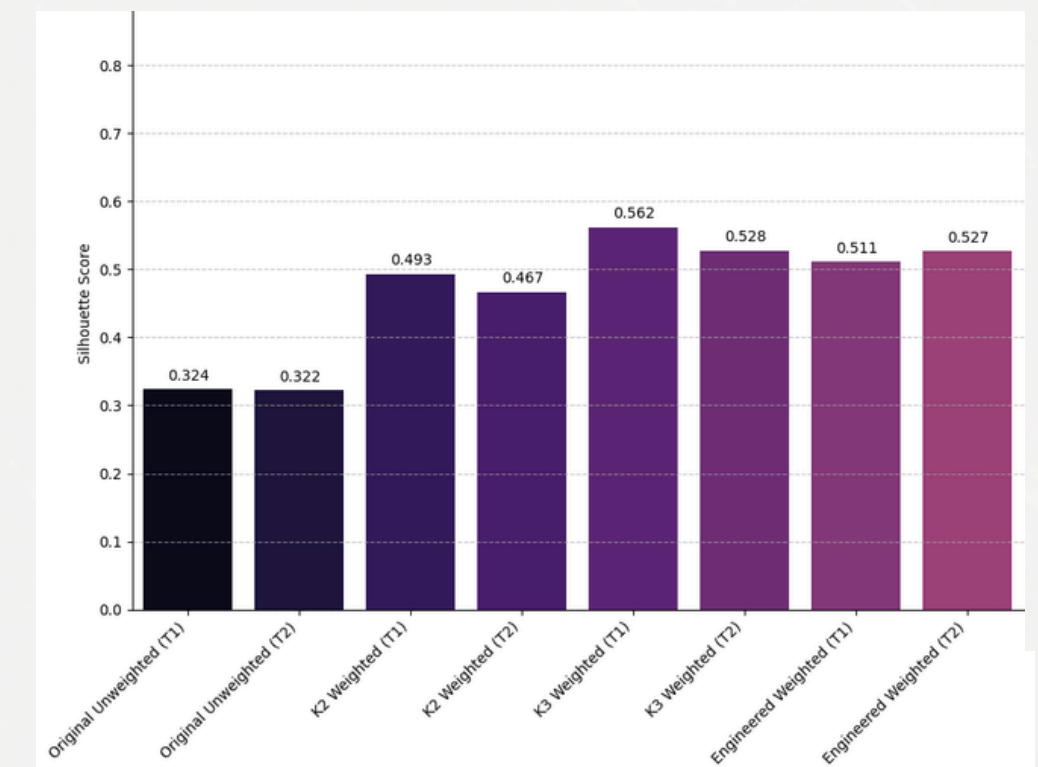
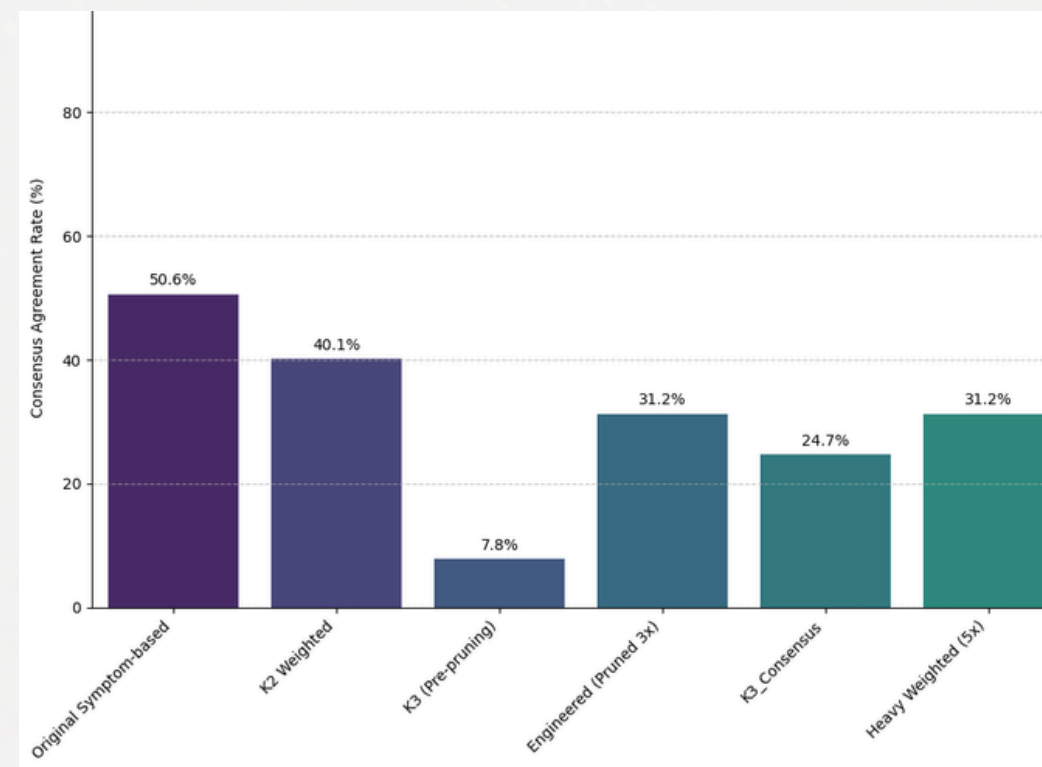
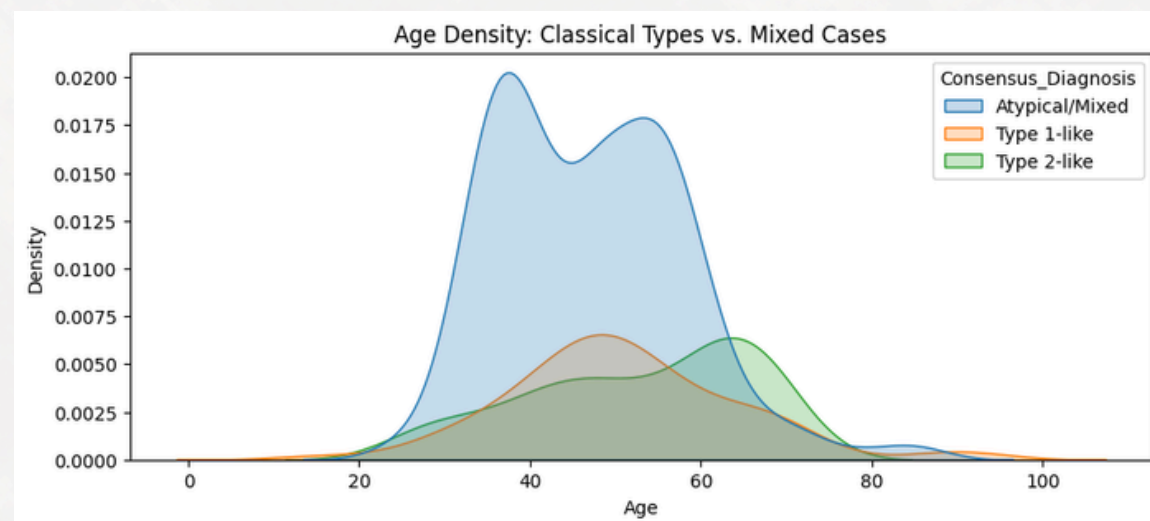
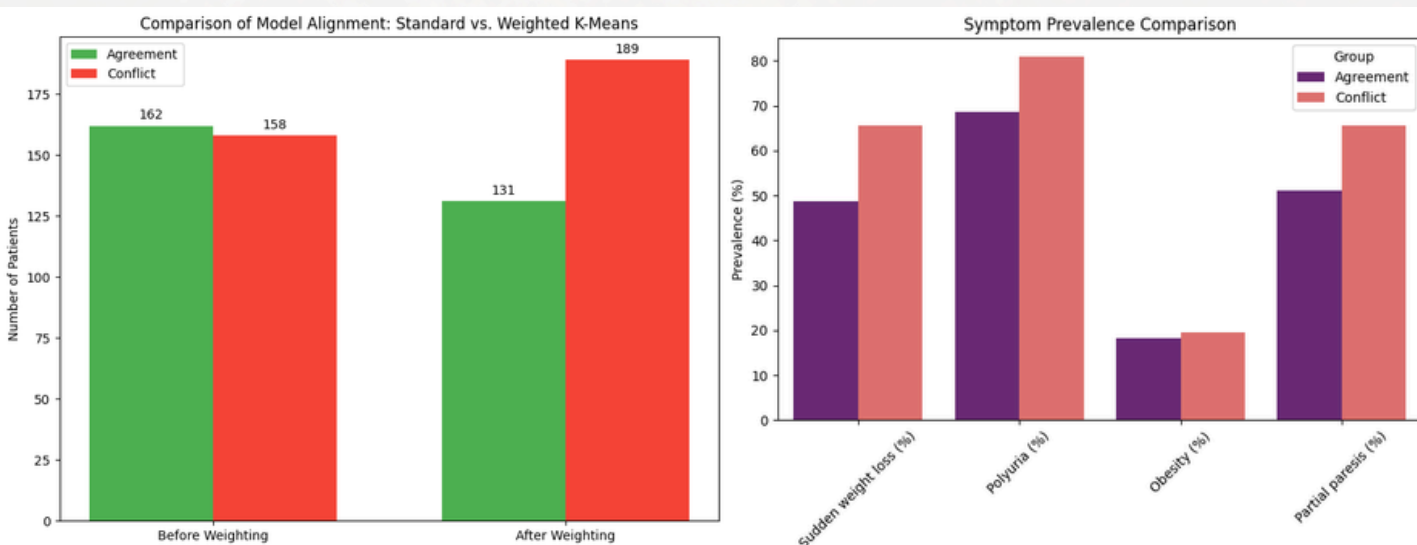
Stage 2: Unsupervised Clustering

- **K-Means** clustering exclusively to the diabetic cohort to uncover latent clinical signatures
- 2 approaches used. The first focuses on type 1 symptoms, and the second on those of type 2
- At the end, a comparison between the 2 approaches is used



Stage 2: Unsupervised Clustering

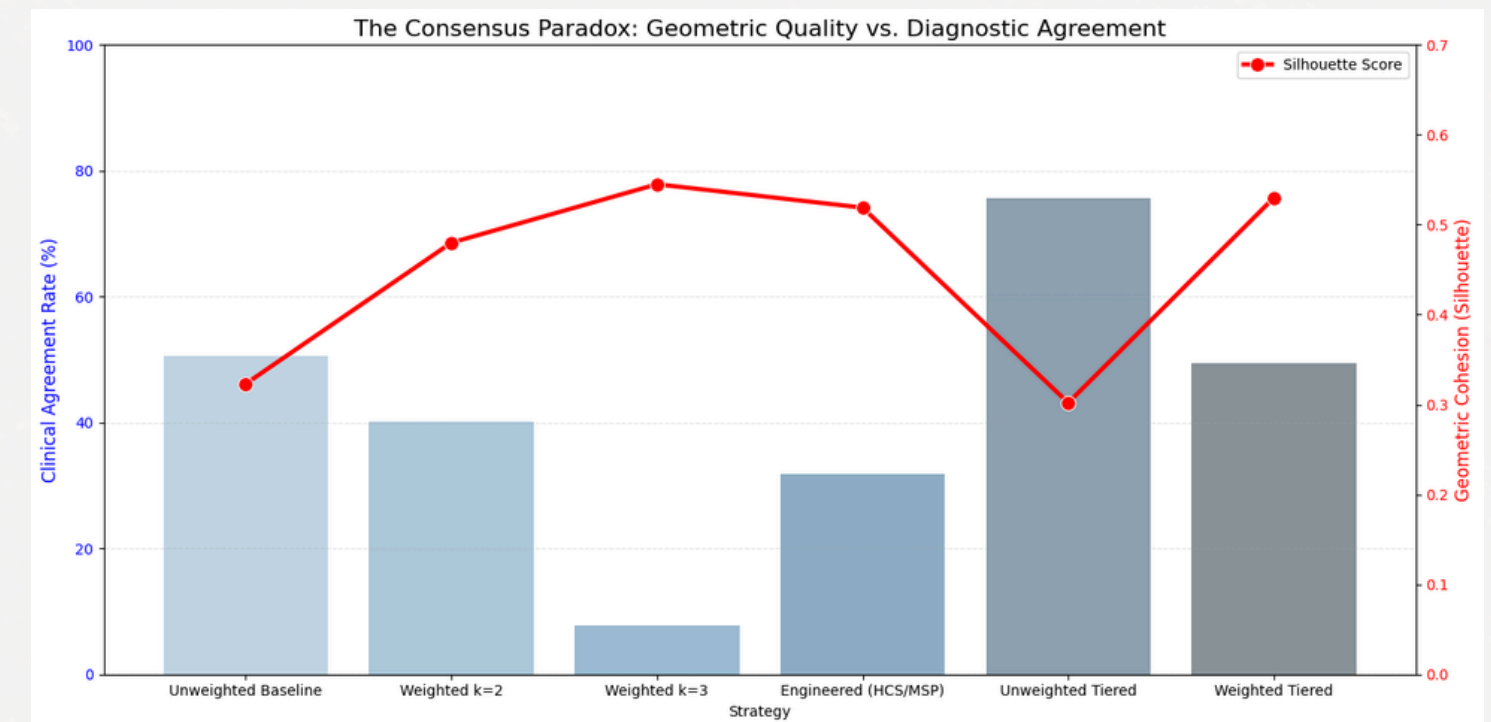
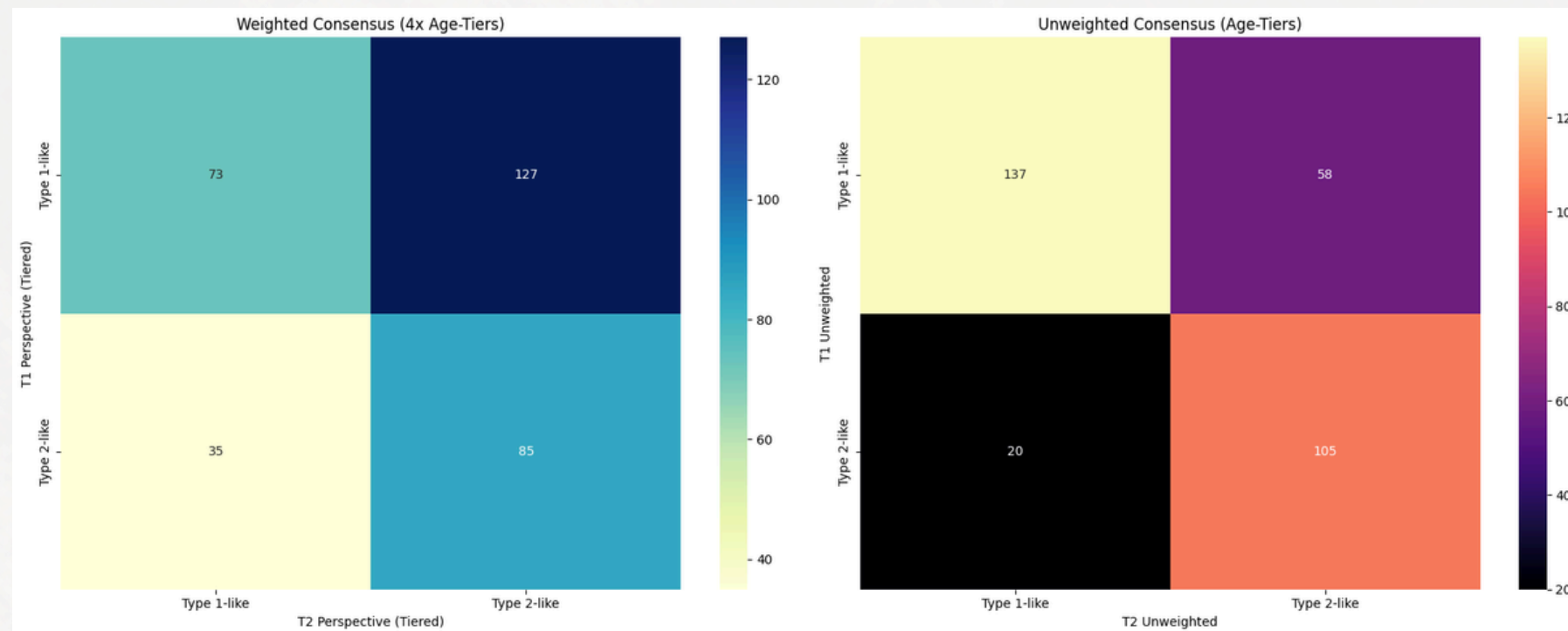
- Introduced a **weighting function**, improving the Silhouette score to 0.49, but reducing consensus
- Consensus Paradox**: Improved geometric separation led to degraded clinical interpretability by attempting to isolate a "mixed" symptom group
- Engineering & Pruning**: With k=3, the Silhouette score jumped to 0.545, but caused diagnostic consensus to collapse to 7.81%



Complex features captured latent noise; pruning low-variance symptoms failed to restore stability

Stage 2: Unsupervised Clustering

- Discretized the cohort into biological age bins to ground clusters in clinical context
- Unweighted Tiered** achieved the highest clinical consensus rate of 75.62%, and **Weighted Tiered** achieved the overall best balance between silhouette score and consensus
- Age-stratification was key to increasing cluster counts for diagnostic triage, without compromising the consensus too harshly



Conclusion

Findings



Domain-driven weighting successfully partitions the cohort, but the "Consensus Paradox" proves that symptom data alone cannot resolve diagnostic ambiguity

The Age-Tiered model provides a robust, interpretable framework for early triage, outperforming high-dimensional complexity

Constraints



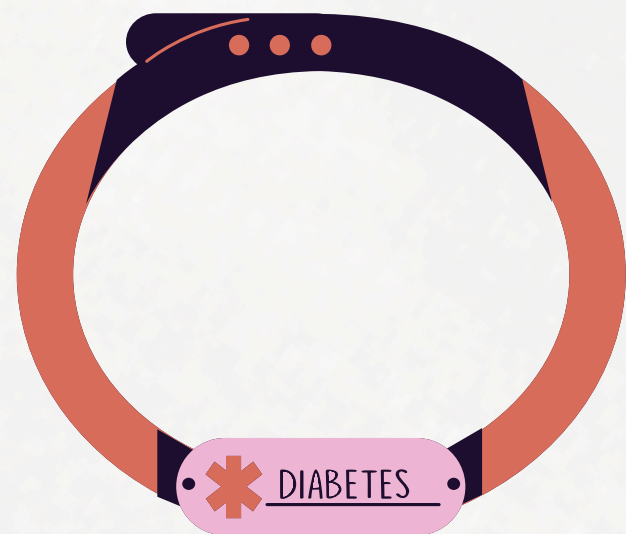
Symptom-based profiles are inherently limited; the existence of "Atypical/Mixed" clusters is a clinical reality, not just a model failure

Future Work



Integrating physiological markers (e.g., HbA1c, BMI) is the essential next step to bridge the gap from symptom-based proxies to definitive clinical diagnosis, as differentiating between metabolic insulin resistance and autoimmune-driven deficiency is key

Thank you



STUDENT
Filippo Velli

